

Yunfan Gao

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Google Scholar: <https://scholar.google.com/citations?&user=iR36jUoAAAAJ>

PROFESSIONAL EXPERIENCE

Jul 2026–Present	Digital Futures Postdoctoral Fellow KTH Royal Institute of Technology • <i>Advisor</i>: Prof. Dr. Danica Kragic Jensfelt • Research on robotic manipulation of dynamic and deformable objects.	Stockholm, Sweden
Mar 2022–Sep 2025	Industrial PhD student in Robotics Bosch Corporate Research • <i>Supervisor</i>: Dr. Niels van Duijkeren • Research and development of real-time-feasible optimal control of mobile robots: (i) enabling smooth navigation in confined space that robots can barely navigate through; (ii) robustifying collision avoidance in the presence of uncertainties caused by plant-model mismatch and other moving objects. • Demonstration of the proposed controllers in real-world robotic experiments. • Extension of the collision-free optimal control methods to manipulation tasks.	Renningen, Germany
Jul 2021–Dec 2021	Software Development Intern Carl Zeiss • Sensor fusion (combining camera and IMU data) for object tracking.	Oberkochen, Germany

EDUCATION

Mar 2022–Jun 2026	PhD in Microsystems Engineering Albert-Ludwigs-Universität Freiburg • <i>Advisor</i>: Prof. Dr. Moritz Diehl • <i>Thesis</i>: Robust model predictive control of robots in confined spaces. • Funded by Bosch Research until Sep 2025; internal PhD student from Oct 2025. • Associated fellow of Marie Skłodowska-Curie Innovative Training Network ELO-X.	Freiburg, Germany
Sep 2019–Jan 2022	Master in Robotics, Systems, and Control ETH Zürich • <i>Thesis project</i>: Projection-based augmented reality with an ANYmal robot, supervised by Dr. Ryan Luke Johns, Perry Franklin, and Prof. Dr. Marco Hutter. • <i>Semester project</i>: Multi-sensor fusion for drone localization, supervised by Dr. David Hug, Dr. Marco Karrer, and Prof. Dr. Margarita Chli. • <i>Award</i>: Degree awarded “with distinction”, Swiss Robotics Master Award	Zürich, Switzerland <i>GPA: 5.82/6.00</i>
Sep 2015–Jun 2019	Bachelor in Electronic Engineering Fudan University • <i>Thesis project</i>: Channel-state-information-based indoor smartphone localization, supervised by Prof. Dr. Yuedong Xu.	Shanghai, China <i>GPA: 3.79/4.00</i>
Sep 2017–Dec 2017	Exchange Program University of California, Santa Barbara	Santa Barbara, the United States <i>GPA: 4.00/4.00</i>

PUBLICATIONS

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| 2026 | <ul style="list-style-type: none"> • Y. Gao, F. Messerer, N. van Duijkeren, R. Dabir, and M. Diehl, “Semi-Infinite Programming for Collision-Avoidance in Optimal and Model Predictive Control,” <i>IEEE Transactions on Robotics</i>, vol. 42, pp. 2399–2418, Apr 2026. [doi][pdf][code] • F. Messerer, Y. Gao, J. Frey, and M. Diehl, “Riccati-ZORO: An efficient algorithm for heuristic online optimization of internal feedback laws in robust and stochastic model predictive control,” <i>European Journal of Control</i>, Jul 2026. [doi][pdf][code] • Y. Gao, “Robust model predictive control of robots in confined spaces,” <i>PhD thesis</i>, Albert-Ludwigs-Universität Freiburg, Jul 2026. [pdf] |
| 2024 | <ul style="list-style-type: none"> • Y. Gao, F. Messerer, N. van Duijkeren, B. Houska, and M. Diehl, “Real-Time-Feasible Collision-Free Motion Planning For Ellipsoidal Objects,” in <i>Proc. of the IEEE Conf. on Decision and Control (CDC)</i>, Dec 2024. [pdf][code] • Y. Gao, F. Messerer, N. van Duijkeren, and M. Diehl, “Stochastic Model Predictive Control with Optimal Linear Feedback for Mobile Robots in Dynamic Environments,” <i>IFAC-PapersOnLine</i>, Aug 2024. [pdf] • J. Frey, Y. Gao, F. Messerer, A. Lahr, M. Zeilinger, and M. Diehl, “Efficient Zero-Order Robust Optimization for Real-Time Model Predictive Control with acados,” in <i>Proc. of the European Control Conf. (ECC)</i>, Jun 2024. [pdf][code] |
| 2023 | <ul style="list-style-type: none"> • Y. Gao, F. Messerer, J. Frey, N. van Duijkeren, and M. Diehl, “Collision-free motion planning for mobile robots by zero-order robust optimization-based MPC,” in <i>Proc. of the European Control Conf. (ECC)</i>, Jun 2023. [pdf] • Y. Gao, F. Messerer, N. van Duijkeren, and M. Diehl, “Optimization-based collision checking between objects represented by Minkowski sums of ellipsoids,” filed at German patent office, 2023. |
| 2021 | <ul style="list-style-type: none"> • Z. Gao, A. Li, Y. Gao, B. Li, Y. Wang, and Y. Chen, “FedSwap: A Federated Learning based 5G Decentralized Dynamic Spectrum Access System,” in <i>IEEE/ACM Int. Conf. on Computer Aided Design (ICCAD)</i>, Nov 2021. [pdf] • Z. Gao, A. Li, Y. Gao, Y. Wang, and Y. Chen, “Hermes: Decentralized Dynamic Spectrum Access System for Massive Devices Deployment in 5G,” in <i>Proc. of the Int. Conf. on Embedded Wireless Systems and Networks (EWSN)</i>, Apr 2021. [pdf] |
| 2020 | <ul style="list-style-type: none"> • Z. Gao*, Y. Gao*, S. Wang, D. Li, and Y. Xu, “CRISLoc: Reconstructable CSI Fingerprinting for Indoor Smartphone Localization,” <i>IEEE Internet of Things Journal</i>, Sep 2020. [pdf] (* These authors contributed equally.) |

TEACHING EXPERIENCE

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| Apr 2026–Jul 2026 | Instructor of Race Car Control Lab
Albert-Ludwigs-Universität Freiburg | Freiburg, Germany |
| | <ul style="list-style-type: none"> • Project-based lab on control and estimation algorithms for model race cars. | |
| Oct 2022–Jun 2023 | Supervision of Master Thesis
Bosch Corporate Research | Renningen, Germany |
| | <ul style="list-style-type: none"> • <i>Thesis project</i>: Optimization-based motion planning using signed-distance maps. | |
| May 2022–Oct 2022 | Supervision of Master Thesis
Bosch Corporate Research | Renningen, Germany |
| | <ul style="list-style-type: none"> • <i>Thesis project</i>: Safety certification of motion control for mobile robots. | |
| 2021 | Teaching Assistant
ETH Zürich | Zurich, Switzerland |
| | <ul style="list-style-type: none"> • Taught tutorial and exercise sessions for the course <i>Programming for Robotics—ROS</i>. | |

PROFESSIONAL SERVICE

Mar 2025–Apr 2025 | Co-organized the Hackathon at the workshop *Future PhD in control*, supported by IEEE CSS and EUCA.

Apr 2023–Oct 2023 | Co-organized the Bosch PhD Conference, attended by over 150 participants.